HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY – VNU-HCM
FACULTY OF COMPUTER SCIENCE AND ENGINEERING



Course: CO5177 – Programming Foundations for Data Analysis and Visualization

Semester 261, Academic Year 2026–2027

PROJECT REPORT

Tabular Data Analysis

Tabular

Group: DataInsight
Lecturer: Lê Thành Sách
Landing page: https://trthminh.github.io/CO5177/

Name	Student ID	Project
Trương Thanh Minh	2580893	Time series
Tô Anh Phát	2580894	Text
Trần Lê Hoàng Long	2570443	Tabular

Ho Chi Minh City, October 2026

Contents

1 Tabular Data Analysis **1**

1.1 Problem and Objectives 1
1.2 Dataset 1
1.3 Data Quality Checks 2
1.4 Exploratory Data Analysis 3
1.4.1 Question 1: Are Bookings Made Earlier More Likely to Be Cancelled? 3
1.4.2 Question 2: Why Are Non Refund Bookings Cancelled Most Often? . . 3
1.4.3 Question 3: Do Special Requests and Repeated Guests Reduce Cancellations? 4
1.5 Data Preprocessing 5
1.5.1 Pipeline and Data Leakage Prevention 5
1.5.2 Encoding and Scaling 5
1.5.3 Comparing Missing-Value Strategies 5
1.5.4 Outlier Handling 6
1.6 Models and Experimental Setup 6
1.6.1 Leakage Experiment 6
1.6.2 Handling Class Imbalance 7
1.7 Results and Evaluation 7
1.8 Question-Driven Figure 9
1.9 Limitations and Future Work 10
1.10 References 10

Chapter 1

Tabular Data Analysis

1.1 Problem and Objectives

When a booking is cancelled close to the arrival date, the hotel usually cannot sell the room again. If the hotel knows in advance which bookings are likely to be cancelled, it can call the guest to confirm, ask for a deposit or accept a few extra bookings to fill the gap. This chapter predicts whether a hotel booking will be cancelled (`is_canceled` = 1) using only the information that is available when the booking is made, such as lead time, length of stay, number of guests, sales channel, deposit type, price and the guest's history.

Since only part of the bookings are cancelled, models are compared mainly by the PR-AUC (average precision) and the F1-score of the cancelled class on the validation set. Accuracy is reported only for reference.

1.2 Dataset

- Name: Hotel Booking Demand (TidyTuesday 2020-02-11 version, identical to the Kaggle copy).
- URL: https://github.com/rfordatascience/tidytuesday/tree/master/data/2020/2020-02-11
- Source: property management systems of two hotels in Portugal, a resort hotel in the Algarve (H1, 40,060 bookings) and a city hotel in Lisbon (H2, 79,330 bookings), arrivals from July 2015 to August 2017.
- Size: 119,390 rows, 32 columns; one row is one booking.
- Target: `is_canceled` (37.0% cancelled in the raw data).
- Missing values: 129,425 cells, in `company` (94.3%), `agent` (13.7%), `country` (0.4%) and `children` (4 cells).
- License / citation: N. António, A. de Almeida, L. Nunes, *Hotel booking demand datasets*, Data in Brief 22 (2019) 41–49, CC BY 4.0.

Each column was checked to see whether it is known at the time of booking. Four columns are only recorded later and were removed: `reservation_status` and `reservation_status_date` (they are the outcome itself), `assigned_room_type` (known at check-in) and `booking_changes` (counted until check-in or cancellation). `arrival_date_year` was also removed because it only covers 2015–2017. The remaining 26 features consist of 15 numeric and 11 categorical columns; `agent` and `company` are stored as numbers but are treated as categorical because they are IDs.

1.3 Data Quality Checks

Table 1.1: Data quality issues and how they were handled

Issue	Size	Handling
Missing `company`, `agent`	94.3% and 13.7% of rows	Kept as a separate `"missing"` category, plus `has_company`, `has_agent` flags
Missing `country`, `children`	488 and 4 cells	Same category approach; `children` filled with the train median
Exact duplicate rows	31,994 rows (26.8%); 34,050 after removing the dropped columns	Removed before splitting
Negative or extreme `adr`	1 row at -6.38, 1 row at 5,400	Removed (valid range 0–1,000)
Bookings with no guests	180 rows	Removed with a multi-column rule
`meal = Undefined`	1,169 rows	Merged with `SC` (no meal package)
Long right tails (`lead_time`, `adr`, stays)	up to 737 days, 50 nights	Kept; `log1p` and `RobustScaler`

Missing values are not random in this dataset. Bookings without a company were cancelled 38.2% of the time, compared with 17.5% for bookings with a company, while bookings without an agent were cancelled 24.7% of the time against 39.0% for those with an agent. An empty `agent` or `company` simply means that the booking was not made through an agency or a company, so these cells were kept as information instead of being filled in.

The duplicates deserve attention. The dataset has no booking ID, so several identical rooms booked by one group become identical rows. The duplicate copies are very different from the rest: 63.1% of them were cancelled (27.5% for the other rows), 42.3% are Non Refund (1.2%) and 46.5% belong to the Groups segment (5.7%). In fact 13,549 of the 14,587 Non Refund bookings are duplicates. Keeping them would place the same booking in both training and test data, so they were removed. After cleaning, 85,175 bookings remain and the cancellation rate drops from 37.0% to 27.7%.

The IQR rule was not useful for the count columns. For `adults`, $Q_1 = Q_3 = 2$, so every booking with a different number of adults was flagged (24.9% of rows). The extreme values that remain, such as 55 adults in a group booking or stays of more than 30 nights, are plausible, so they were kept.

1.4 Exploratory Data Analysis

Several patterns already appear before modelling. No booking with a parking request was cancelled (7,416 bookings), guests from Portugal cancelled 56.6% of their bookings compared with 16.7–20.2% for guests from Germany, France and the UK, and the City Hotel had a higher cancellation rate (41.7%) than the Resort Hotel (27.8%). Because the duplicates are concentrated in some groups, the three questions below are computed both on all rows and on the de-duplicated rows that are later used for modelling.

1.4.1 Question 1: Are Bookings Made Earlier More Likely to Be Cancelled?

On de-duplicated rows, the cancellation rate of the City Hotel rises from 10.5% for bookings made at most 7 days ahead to 45.7% for bookings made 181–365 days ahead. The Resort Hotel follows the same trend up to one year (6.0% to 35.2%). The duplicates inflate the long lead times: for the City Hotel the 181–365 day group reaches 62.7% on all rows.

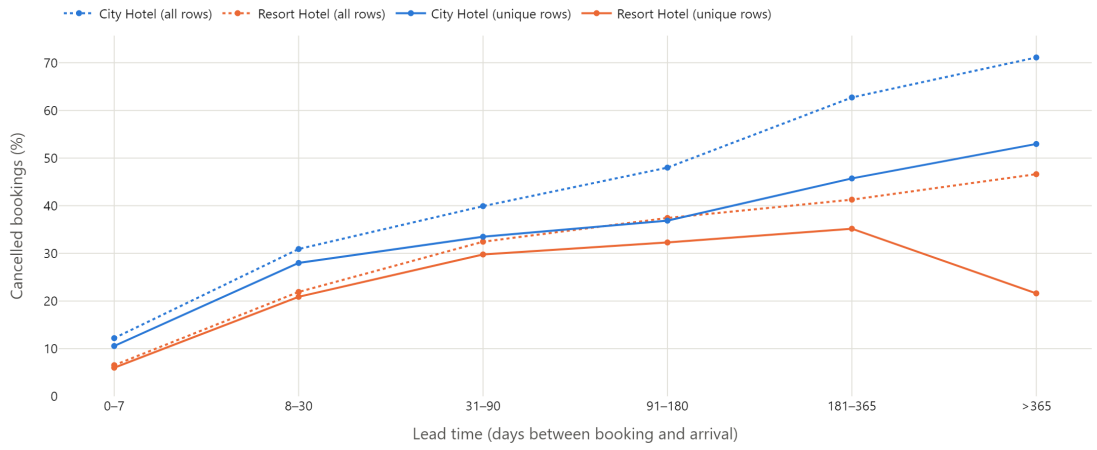


Figure 1.1: The cancellation rate increases with lead time: 10.5% at $\leq$ 7 days and 45.7% at 181–365 days for the City Hotel (solid lines: de-duplicated rows, dotted lines: all rows).

1.4.2 Question 2: Why Are Non Refund Bookings Cancelled Most Often?

Non Refund bookings are cancelled 99.4% of the time on all rows and still 95.0% on de-duplicated rows, but only 992 unique Non Refund bookings remain. Compared with bookings without deposit, they are mostly group or tour operator bookings (62.8% Groups and 28.2% Offline TA/TO), almost all from Portugal (94.9%), and made 192 days ahead in median instead of 47. A likely explanation is that tour operators reserve blocks of rooms early and release them

when they cannot sell them, so the non-refundable condition belongs to the agency contract rather than to an individual guest. The data does not contain a column that confirms this.

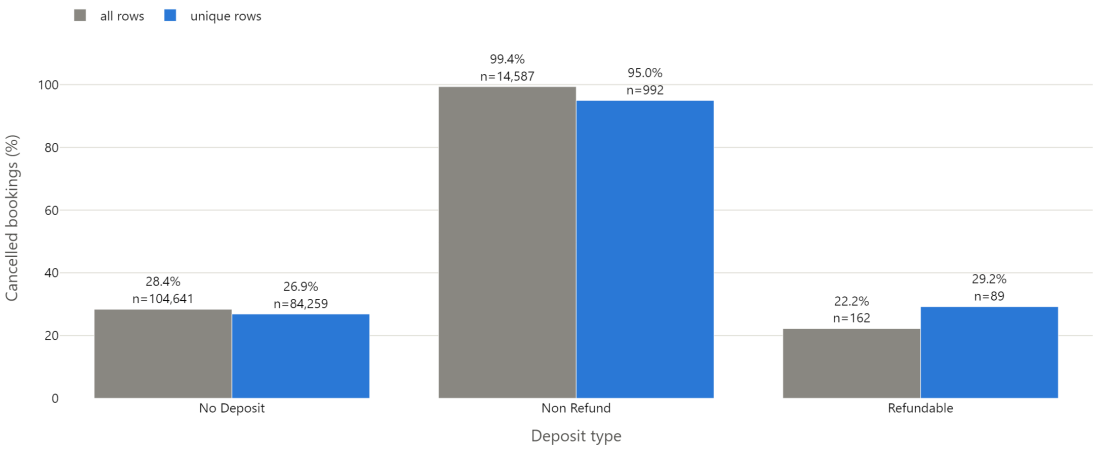


Figure 1.2: Non Refund bookings are still cancelled 95.0% of the time after removing duplicates, but only 992 of the 14,587 Non Refund rows are unique.

1.4.3 Question 3: Do Special Requests and Repeated Guests Reduce Cancellations?

For new guests, the cancellation rate falls from 34.7% without any special request to 23.2% with one request and 16.9% with three or more. Repeated guests cancel only 7.7% of their bookings against 28.5% for new guests, and they cancel less at every number of requests.

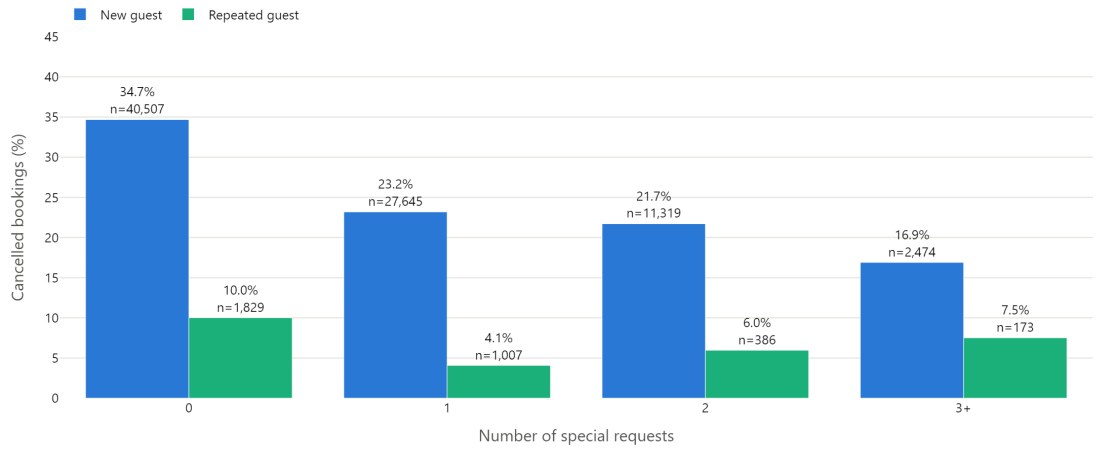


Figure 1.3: One special request lowers the cancellation rate of new guests from 34.7% to 23.2%; repeated guests cancel 7.7% compared with 28.5% for new guests.

1.5 Data Preprocessing

1.5.1 Pipeline and Data Leakage Prevention

Row-wise rules (normalising text, removing duplicates and impossible values, creating `total_nights`, `total_guests`, `has_agent` and `has_company`) were applied before the split. All statistics (medians, scaling parameters, category lists) are learned inside a `Pipeline` with a `ColumnTransformer`, so they only come from the training data, including inside cross-validation.

Table 1.2: Stratified train / validation / test split (70/15/15)

Set	Rows	Cancelled
Train	59,622	27.7%
Validation	12,776	27.7%
Test	12,777	27.7%

1.5.2 Encoding and Scaling

Numeric columns are filled with the training median and scaled with `RobustScaler`, which uses the median and IQR and is less affected by the long tails. Nominal columns are one-hot encoded, and categories that cover less than 1% of the training rows are merged into one column, which keeps `country`, `agent` and `company` manageable. The arrival month is encoded with `OrdinalEncoder` in calendar order because it has a natural order. The final preprocessing produces 81 columns.

1.5.3 Comparing Missing-Value Strategies

Four strategies were compared with the same logistic regression model (`class_weight="balanced"`) on the validation set.

Table 1.3: Missing-value strategies (validation, cancelled class)

Strategy	Train rows	F1	PR-AUC	ROC-AUC
(a) Drop incomplete rows	122	0.184	0.289	0.524
(b) Drop `company`, median/mode	59,622	0.645	0.655	0.846
(c) Missing as a category + flags	59,622	0.647	0.654	0.849
(d) KNN imputer + category	59,622	0.647	0.654	0.849

Dropping rows is not an option because almost every row lacks `company`, leaving only 122 training rows. The other three strategies give almost the same scores. Strategy (c) was chosen because it does not drop any column and keeps the meaning of an empty cell (no agency or company). KNN gives exactly the same result as (c) because only four numeric cells are missing.

1.5.4 Outlier Handling

Three ways of handling the long-tailed continuous columns were compared with the same model. `log1p` improved PR-AUC from 0.654 to 0.674 and F1 from 0.647 to 0.667, while clipping at the 1st and 99th percentiles changed almost nothing (PR-AUC 0.654) because only 0.3–1% of the rows are clipped. `log1p` was therefore kept. It only helps linear models, since tree models depend on the order of values only.

1.6 Models and Experimental Setup

The compared models are a majority-class baseline (`DummyClassifier`), logistic regression, a decision tree, a random forest and histogram gradient boosting, each with and without `class_weight="balanced"` where available. All models are fitted on the training set and compared on the validation set. The best family was then used to compare ways of handling class imbalance and to tune hyperparameters with `RandomizedSearchCV` (10 combinations, 3-fold stratified cross-validation on the training set, scored by average precision).

Table 1.4: Comparison of models on the validation set (cancelled class)

Model	Precision	Recall	F1	PR-AUC	Accuracy
Baseline (majority class)	0.000	0.000	0.000	0.277	0.723
Logistic Regression	0.679	0.531	0.596	0.683	0.801
Logistic Regression (balanced)	0.562	0.822	0.667	0.674	0.773
Decision Tree	0.719	0.576	0.640	0.701	0.820
Random Forest	0.741	0.648	0.692	0.775	0.840
HistGradientBoosting	0.733	0.637	0.682	0.779	0.835
HistGradientBoosting (balanced)	0.602	0.856	0.707	0.775	0.803

Tree ensembles clearly outperform logistic regression (PR-AUC around 0.78 against 0.67–0.68), which matches the non-linear relationships seen in the EDA. Histogram gradient boosting has the highest PR-AUC and trains much faster than the random forest, so it was used for the next steps.

1.6.1 Leakage Experiment

To show why the post-booking columns were removed, the same gradient boosting model was trained again with them added back.

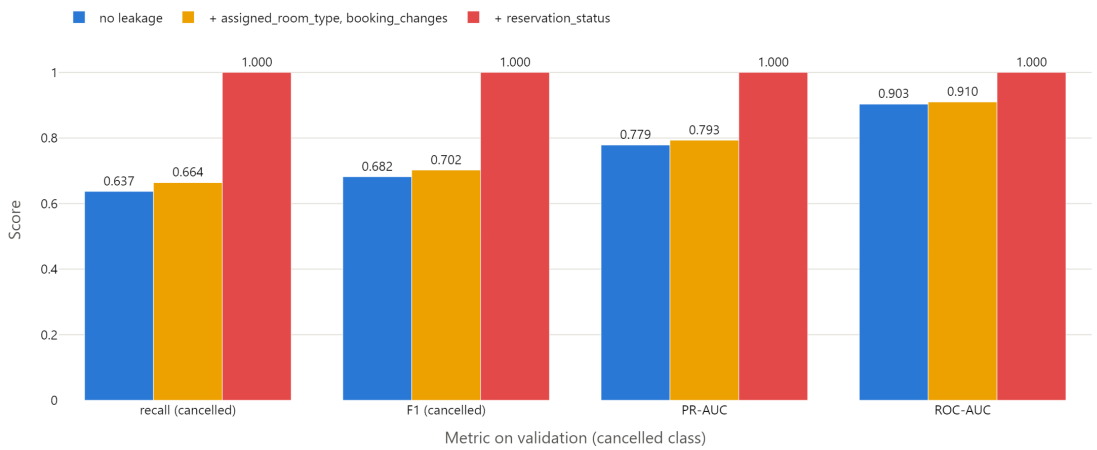


Figure 1.4: Adding check-in information raises PR-AUC from 0.779 to 0.793, and adding `reservation_status` gives a perfect but meaningless 1.000.

With `reservation_status` every metric equals 1.000, because the column is the outcome itself. The smaller increase from `assigned_room_type` and `booking_changes` is harder to notice and could easily be mistaken for a real improvement, but these columns do not exist when the prediction has to be made.

1.6.2 Handling Class Imbalance

Table 1.5: Imbalance handling with HistGradientBoosting (validation, cancelled class)

Method	Precision	Recall	F1	PR-AUC
No handling	0.733	0.637	0.682	0.779
`class_weight="balanced"`	0.602	0.856	0.707	0.775
SMOTE (train only)	0.688	0.712	0.700	0.770
Tuned threshold (0.374)	0.666	0.770	0.715	0.779

All three methods increase the recall of cancellations. Tuning the decision threshold with cross-validation on the training set gives the best F1 and keeps the PR-AUC unchanged, since the model itself does not change. Class weighting gives the highest recall but its precision drops to 0.602, and SMOTE has the lowest PR-AUC, probably because it interpolates one-hot columns and creates bookings that cannot exist.

1.7 Results and Evaluation

The final model is the tuned gradient boosting model (learning rate 0.1, 200 iterations, 31 leaf nodes, at least 100 samples per leaf, L2 regularisation 1.0) with a decision threshold of 0.364. It was refitted on the training and validation sets (72,398 rows) and evaluated once on the test set.

Table 1.6: Final model on the validation and test sets (cancelled class)

Model	Set	Precision	Recall	F1	PR-AUC	ROC-AUC	Accuracy
Baseline	test	0.000	0.000	0.000	0.277	0.500	0.723
HGB tuned + threshold	validation	0.660	0.782	0.715	0.783	0.905	0.828
HGB tuned + threshold	test	0.649	0.775	0.707	0.775	0.900	0.822

The test scores are within 0.011 of the validation scores, so there is no sign of overfitting to the validation set. Compared with the baseline, PR-AUC rises from 0.277 to 0.775 and F1 from 0 to 0.707, while accuracy only goes from 72.3% to 82.2%. Out of 3,537 cancelled bookings in the test set the model detects 2,741 (77.5%), and 1,481 of the 9,240 kept bookings (16.0%) are flagged by mistake.

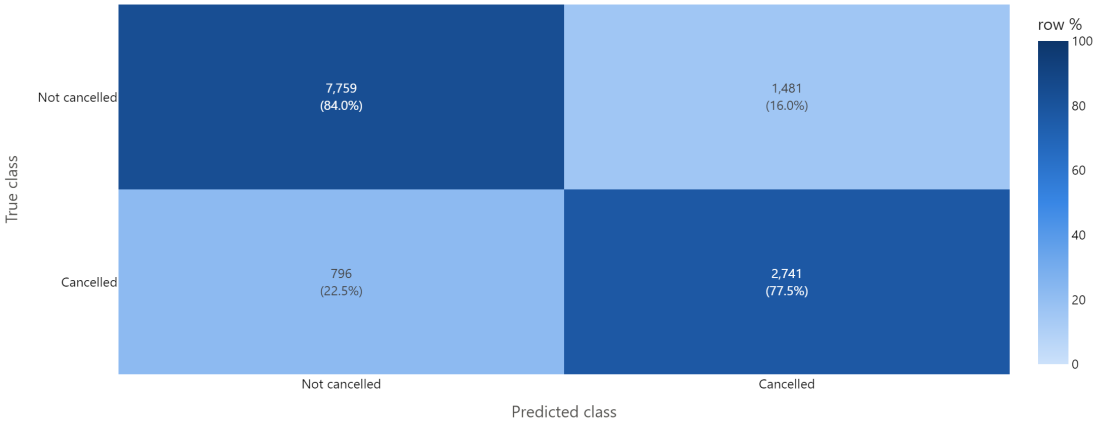


Figure 1.5: On the test set the model detects 77.5% of cancellations and wrongly flags 16.0% of the bookings that were kept.

The permutation importance on the test set agrees with the EDA. Shuffling `country` lowers PR-AUC by 0.162, followed by `lead_time` (0.127), `agent` (0.115), `total_of_special_requests` (0.108) and `market_segment` (0.082). `deposit_type` only ranks ninth (0.042) because few Non Refund bookings remain after removing duplicates.

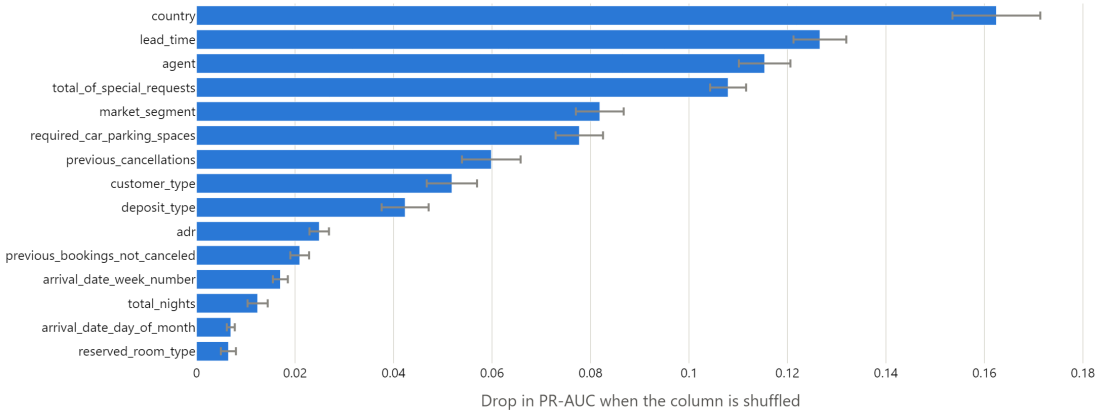


Figure 1.6: Country, lead time, agent and the number of special requests are the most important features, in line with the EDA.

The 796 missed cancellations look like low-risk bookings: their median lead time is 35 days (94 days for the detected cancellations) and 66.3% of them have a special request (32.0%). Their reasons for cancelling, such as a change of plans or illness, are not recorded in the booking data, which limits what any model can learn from it.

1.8 Question-Driven Figure

Question: Does handling class imbalance improve the recall of cancelled bookings?

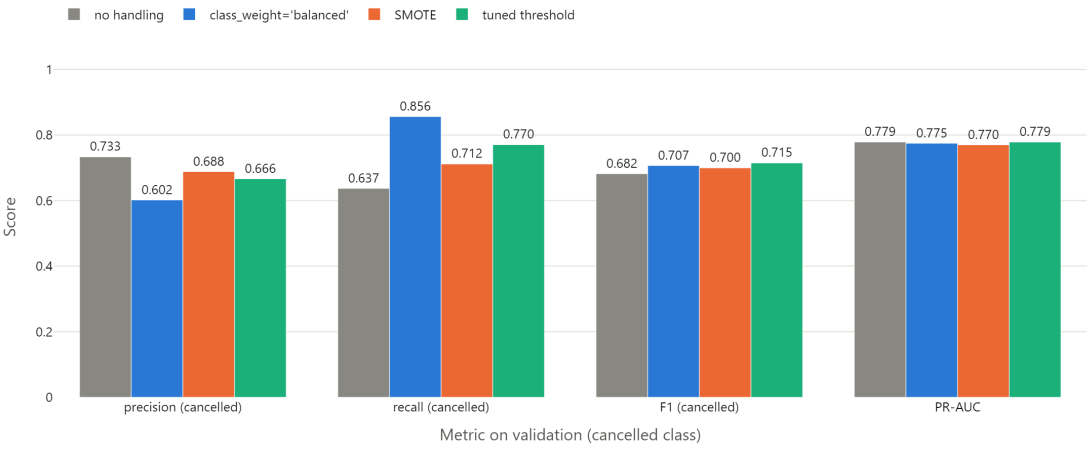


Figure 1.7: Handling imbalance raises the recall of cancellations from 0.637 to 0.856 with class weights and to 0.770 with a tuned threshold; the tuned threshold gives the best F1 (0.715) without lowering PR-AUC.

Yes. Without any handling, the model misses 36.3% of the cancelled bookings in the validation set. With a threshold of 0.374 it misses 23.0% and still keeps a precision of 0.666. For a hotel, a false alarm usually costs a confirmation call, while a missed cancellation can leave a room empty, so trading some precision for recall is reasonable here.

1.9 Limitations and Future Work

- The data covers only two hotels in Portugal with arrivals from 2015 to 2017, so the results may not transfer to other hotels or to the period after COVID.
- Without a booking ID, duplicated entries cannot be separated from rooms of the same group. Removing duplicates changes the cancellation rate from 37.0% to 27.7%.
- `country` may partly be updated at check-in. If so, cancelled bookings would more often have an empty or default country, which could explain part of its importance. This could not be verified with the available data.
- The split is random rather than chronological, so the test score may be optimistic for predicting future bookings.
- Future work: a chronological split (train on 2015–2016, test on 2017), a check of `country` by removing it, a threshold chosen from the actual costs of empty rooms and confirmation calls, and SMOTENC instead of SMOTE.

1.10 References

- N. António, A. de Almeida, L. Nunes, *Hotel booking demand datasets*, Data in Brief 22 (2019) 41–49. CC BY 4.0.
- R for Data Science Online Learning Community, *TidyTuesday 2020-02-11: Hotels*, https://github.com/rfordatascience/tidytuesday.
- F. Pedregosa et al., *Scikit-learn: Machine Learning in Python*, JMLR 12 (2011) 2825–2830.
- G. Lemaître, F. Nogueira, C. K. Aridas, *Imbalanced-learn: A Python Toolbox to Tackle the Curse of Imbalanced Datasets in Machine Learning*, JMLR 18 (2017) 1–5.